

S^n , ~~S^1~~ , RP^n , $\langle \dots \rangle$
 $(n \geq 1)$
 $\downarrow$
 $\pi_1 = 0$, 2-dim mfd, $L(p,q)$

$M \cup N$, $\wedge \vee$, $\langle X, S \rangle / \langle e^{2\pi i p}, e^{2\pi i q} \rangle$
 π_0, π_1^2 , Kleim., free product, $ababab...$
 $\mathbb{Z} \times \mathbb{Z}$, $\langle a, b \mid aba = b \rangle$, $ababab^t = e$
 $ab = ba \rightarrow$ group $\mathbb{Z} \times \mathbb{Z}$, $\langle a, b \mid ab = ba \rangle$
 $S^1 \vee S^2$, $T: \mathbb{Z} \times \mathbb{Z}$, $T \hookrightarrow \mathbb{Z} \times \mathbb{Z}$, $\mathbb{Z} \hookrightarrow \mathbb{Z} \times \mathbb{Z}$ theorem

Theorem: $M: n$ -dim with $(n \geq 2)$
 then $\pi_1(M) \cong \pi_1(M \setminus \{x\})$
 $\pi_1(RP^n) \cong \mathbb{Z}_2$ ($n \geq 2$), $\pi_1(S^n) = 0$ ($n \geq 2$)
 $\pi_1(M) \cong \pi_1(M \setminus \{x\})$

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Example: $S^2/p = S$, S^1 : 0-cell $\hookrightarrow$, 1-cell $\hookrightarrow$
 S^2 : 0-cell $\hookrightarrow$, 2-cell $\hookrightarrow$
 RP^2 : 0-cell: $\hookrightarrow$, 1-cell: $\hookrightarrow$, 2-cell: $\hookrightarrow$
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$RP^2: \mathbb{Z}_2$, $\langle a \mid a^2 \rangle$, $a^2 = 1$
 RP^2 : 0-cell $\hookrightarrow$, 1-cell $\hookrightarrow$, 2-cell $\hookrightarrow$
 $\pi_1(RP^2) = \pi_1(RP^2) = \mathbb{Z}_2$
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$\mathbb{R}/\mathbb{Z}$, $\langle \dots \rangle$, $\langle \dots \rangle$, $\langle \dots \rangle$
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